SIH PROBLEM STATEMENT INITIAL APPROACH

27/08/26

1. Introduction

This document explains the initial approach for building a minimal, working prototype of an AI-driven social-media analytics framework. The strategy focuses on proving the core concept rather than attempting to implement the complete SIH solution in the internal round.

2. What the Problem Statement Is Asking

The problem asks for an AI-driven social-media analytics framework that takes raw social-media data and converts it into meaningful audience insights. The proposed understanding is based on four core analytical questions and one supporting requirement.

Question	Analysis Area	Purpose
What are people saying?	Trend / Topic Detection	Identify topics, hashtags and frequently discussed terms.
How are people feeling?	Sentiment & Emotion Analysis	Understand positive, negative and neutral reactions.
Who is the audience?	Demographic Profiling	Understand audience characteristics.
How does information spread?	Network / Link Analysis	Study interactions and information flow.
Supporting requirement	Continuous Data Collection & Timeline Management	Collect posts/interactions and preserve timestamps for analysis over time.

3. Overall Concept

The basic concept can be represented as:

Collect social-media data → Organise it chronologically → Analyse content, sentiment, audience and interactions → Present actionable insights

4. Full Vision vs. Internal Round Goal

A complete SIH solution could theoretically support multiple platforms feeding into a unified pipeline. However, the initial approach recommends not attempting to build the complete system for the internal round.

The Tempting Goal

"Build the complete SIH solution."

The Right Goal

"Build a tiny working demonstration that proves the core concept."

The reason is limited development time. The approach prioritizes depth over breadth: instead of implementing every SIH component partially, the team should implement a smaller set of components properly.

5. Datasets Selected for the Prototype

The proposed prototype uses two existing datasets, Twitter and Instagram, to demonstrate different parts of the broader audience-intelligence concept.

5.1 Twitter – Avengers: Endgame Dataset

The Twitter dataset contains tweet text, timestamps, users, replies, retweets, likes and retweet counts.

This makes it suitable for studying how a conversation around a specific event evolved over time.

Demonstrations Possible with Twitter

- Sentiment Analysis – classify tweets as Positive, Negative or Neutral and show sentiment changes over time.
- Trend / Timeline Analysis – identify popular topics, hashtags and frequently discussed terms and track changes in discussion.
- Basic Network Analysis – use screenName → replyToSN to create a user-interaction graph and identify highly connected users.

5.2 Instagram – Content Performance Dataset

The Instagram dataset contains account information, follower counts, content categories, media types, timestamps, likes, comments, shares, saves, reach, impressions, engagement rate and followers gained.

It is useful for analysing content and engagement performance.

Demonstrations Possible with Instagram

- Identify which content categories perform best.
- Compare performance across different media types.
- Study the relationship between follower count and engagement.
- Analyse likes, comments, shares and saves.
- Analyse reach and impressions.
- Study engagement rate and followers gained.

6. How the Prototype Will Be Built

The proposed architecture is intentionally simple. The initial approach does not require APIs, cloud infrastructure, streaming pipelines or custom ML training. The existing CSV files act as the data source.

Stage	Technology / Source	Role
Data Source	Existing CSV files	Provide the prototype data; no live API or streaming is required.

Processing	Pandas	Load, clean, structure and normalize rows.
Split Streams	Twitter + Instagram	Twitter: sentiment, trends and network; Instagram: engagement and content analysis.
Dashboard	Simple visual summary	Present actionable insights and results.

7. Proposed Processing Flow

1. Load the existing CSV datasets.
2. Clean and structure the data using Pandas.
3. Preserve timestamps so that changes can be analysed over time.
4. Process the Twitter stream for sentiment, trends and basic network analysis.
5. Process the Instagram stream for content and engagement analysis.
6. Combine the results into a simple visual dashboard.
7. Present insights that demonstrate the broader audience-intelligence concept.

8. Prototype Scope

The internal-round prototype should concentrate on a small number of demonstrable capabilities:

- Twitter sentiment classification and sentiment-over-time analysis.
- Twitter trend/topic and timeline analysis.
- Basic Twitter interaction/network analysis.
- Instagram content-performance analysis.
- A simple dashboard that summarizes the results.

9. What the Prototype Demonstrates

Although the prototype is intentionally small, it demonstrates the central idea of the problem statement: social-media data can be collected, organised, analysed from multiple perspectives and converted into understandable audience insights.

Twitter demonstrates conversational, temporal and interaction-based intelligence, while Instagram demonstrates content and engagement intelligence.

10. Limitations of the Initial Approach

- The prototype uses existing CSV files rather than live social-media APIs.
- No streaming or continuous live-data pipeline is required at this stage.
- No custom machine-learning training is planned for the initial prototype.
- Demographic profiling is part of the overall problem but is not identified as a main demonstration in the two-dataset prototype.
- The network component is basic and is based on available reply relationships.
- The prototype is intended to prove the concept, not represent the complete final SIH architecture.

11. Recommended Internal-Round Strategy

- Keep the architecture simple and reliable.
- Use the existing datasets as the immediate data source.
- Focus on a few features that can be demonstrated clearly.
- Prioritize working analysis over a large number of incomplete modules.
- Use the dashboard to communicate insights visually.
- Treat the prototype as a foundation for later expansion into a broader unified system.

12. Conclusion

The initial approach is based on building a small but functional demonstration of AI-driven social-media analytics. Instead of trying to implement every requirement at once, the team can demonstrate the core concept using the Twitter Avengers: Endgame dataset for sentiment, trend/timeline and basic network analysis, and the Instagram dataset for content and engagement analysis.

The proposed architecture remains intentionally lightweight: existing CSV files provide the data, Pandas handles cleaning and structuring, separate analysis streams process Twitter and Instagram, and a simple dashboard presents the results. This approach provides a practical foundation for further development while keeping the internal-round scope manageable.

SOURCE: ..\Introduction - PS & Approach.pdf

DATASET

- 29,999 Instagram post records
- 23 columns/signals
- 13 numeric fields + 9 text/category fields
- Reported missing values: 0
- The dataset is structurally complete.

WHAT DATA IS AVAILABLE

- Account: post_id, account_id, account_type
- Audience: follower_count
- Content: media_type, content_category, caption_length, hashtags_count
- Traffic: traffic_source, has_call_to_action
- Time: post_datetime, post_date, post_hour, day_of_week
- Engagement: likes, comments, shares, saves
- Exposure: reach, impressions
- Growth: followers gained
- Outcome: engagement rate + performance bucket label.

MAIN ANALYTICAL DIRECTION OF THE MEETING

- Likes ⟶ attention/reaction
- Comments ⟶ conversation
- Shares ⟶ propagation
- Saves ⟶ content value
- Reach ⟶ unique exposure
- Impressions ⟶ total exposure.

IMPORTANT ANALYSES PROPOSED

- Which media type performs best
- Which content category performs better

- Relationship between caption length and performance
- Impact of hashtag usage
- Top posts by engagement/reach/shares/saves/follower growth
- Identifying high-performing posts.

TIMING ANALYSIS

- Best posting hour
- Best day of the week
- Peak audience activity
- Peak performance window

TRAFFIC SOURCE + CTA

- Reach
- Engagement rate
- Shares
- Saves
- Followers gained

NLP IS NOT DIRECTLY POSSIBLE YET

- Sentiment analysis
- Keyword extraction
- Topic detection
- Rising narrative detection
- These cannot yet be demonstrated on the actual data. The NLP layer can be added once raw captions are available.

ML / PREDICTION

- Proposed target: performance_bucket_label
- Possible models: Logistic Regression; Decision Tree; Random Forest; Gradient Boosting

- Evaluation: Accuracy; Precision; Recall; F1; Feature importance / explainability.

IMPORTANT TECHNICAL DISCUSSION: DATA LEAKAGE

- If performance_bucket_label is created from engagement_rate, engagement_rate should not be used as an input feature for the model.
- Similarly, likes/comments/shares/saves are available after the post outcome, so these features may create leakage in pre-post prediction.
- Therefore, the team must first define whether the prediction is to be made: before the post is published; shortly after publication; or after the campaign is completed.
- This is a critical decision for the final ML architecture.

PROPOSED FINAL SYSTEM

- Overall pipeline: Instagram CSV/API ⟶ Pandas ⟶ Cleaning & Feature Engineering ⟶ Analytics ⟶ NLP ⟶ ML ⟶ Dashboard
- Expected dashboard outputs: Account performance; Content performance; Sentiment; Topics; Trends; Anomaly alerts.

MEETING OBJECTIVE

- Analyze the Twitter dataset related to the **#AvengersEndgame** conversation.
- Clean and prepare the raw Twitter dataset for analysis.
- Analyze Twitter conversation activity over time.
- Compare **original tweets and retweets**.
- Perform sentiment analysis using the **VADER baseline model**.

DATASET CLEANING AND PREPARATION

- The original Twitter dataset contained **17 columns**.
- The created column was converted into a proper datetime format.
- **5-minute time bins** were created for time-based analysis.
- Separate hour and minute columns were created.
- Is Retweet was standardized to is retweet.
- A tweet_type column was created to classify tweets as:
  - Original
  - Retweet
- A text_clean column was created for cleaned tweet text.
- Irrelevant fields were removed.

- Important engagement-related information was retained.

WORKING DATASET

The following key columns were retained for analysis:

- id
- text
- text_clean
- created
- date
- hour
- minute
- time_bin
- user
- is_retweet
- reply_to_user
- retweet_count
- favorite_count
- tweet_type

TIMELINE ANALYSIS – TOTAL TWEETS

- The Twitter conversation was divided into **5-minute time windows**.
- Two separate streams were analyzed:
    - **Original Tweets** – new content creation.
    - **Retweets** – content sharing and amplification.
- Total tweets were calculated as:

Total Tweets = Original Tweets + Retweets

- This separation helped distinguish **content creation** from **content amplification**.

KEY FINDING

- Retweets dominated the conversation throughout the analyzed period.
- The reported retweet ratio was approximately **0.90**.
- This means roughly **90% of the conversation consisted of retweets** in each time window.

- Original tweets represented only a relatively small and steady portion of the conversation.
- The overall conversation was therefore strongly **sharing-driven**.

TWITTER CONVERSATION ACTIVITY OVER TIME

- Activity analysis focused on **engagement and amplification**, rather than only posting volume.
- The retweet ratio remained approximately **0.90** throughout the observed time period.
- Retweeting was the dominant form of activity.
- The conversation showed strong content-sharing behavior.

ORIGINAL TWEETS VS. RETWEETS

- Original tweets remained **low and relatively steady** over time.
- Retweets were **much higher and more fluctuating**.
- Retweets overwhelmingly dominated the conversation at the observed time points.
- The presentation characterizes the conversation as **retweet-driven and viral**.
- The analysis indicates:
    - High content amplification.
    - Potentially high reach through repeated sharing.
    - A comparatively narrow base of original authorship.

SENTIMENT ANALYSIS – VADER BASELINE

- Sentiment analysis was performed using **VADER**.
- VADER is a **pre-trained sentiment analysis model**.
- It can score tweets without requiring custom model training.
- For each cleaned tweet, VADER provides:
    - Negative intensity score.
    - Neutral intensity score.
    - Positive intensity score.
    - Compound score ranging from **−1 to +1**.

SENTIMENT CLASSIFICATION

Tweets were classified using the compound score:

- **Compound ≥ +0.05 → Positive**
- **Compound ≤ −0.05 → Negative**
- **Between −0.05 and +0.05 → Neutral**
- Each tweet was assigned a sentiment label for downstream analysis.

SENTIMENT DISTRIBUTION – ALL TWEETS

- **Positive:** 52.94%
- **Neutral:** 38.42%
- **Negative:** 8.64%

Observation: The overall sentiment distribution was predominantly positive.

SENTIMENT DISTRIBUTION – ORIGINAL TWEETS ONLY

- **Neutral:** 46.56%
- **Positive:** 38.96%
- **Negative:** 14.48%

Observation: Original tweets contained a higher proportion of neutral and negative sentiment compared with the overall dataset.

SENTIMENT ANALYSIS – KEY FINDINGS

- The overall Twitter conversation was **net-positive**.
- Positive sentiment across all tweets was partly influenced by retweets spreading positive content.
- Original tweets were more neutral.
- Original tweets also contained a noticeably higher proportion of negative sentiment.
- Sentiment remained relatively stable throughout the analyzed time window.
- Negative sentiment was more concentrated in original tweets.
- Positive content was amplified through retweets.

KEY DISCUSSION POINTS

DATA PREPARATION

- Raw Twitter data was cleaned and transformed.
- Datetime information was standardized.
- Time-based features were created.
- Original tweets and retweets were separated.
- Cleaned tweet text was prepared for sentiment analysis.

TIMELINE ANALYSIS

- Activity was analyzed using **5-minute intervals**.
- Original tweets and retweets were analyzed separately.

- Retweets consistently represented the majority of the conversation.
- The approximate retweet ratio was **0.90**.

CONVERSATION ANALYSIS

- The conversation was strongly sharing-driven.
- Retweets played the major role in content amplification.
- Original tweet activity remained comparatively low and steady.
- The observed conversation can be characterized as **retweet-driven and viral**.

SENTIMENT ANALYSIS

- VADER was used as the baseline sentiment model.
- Tweets were classified as positive, neutral, or negative.
- Overall sentiment was net-positive.
- Original tweets showed a comparatively higher negative sentiment share.

OVERALL MEETING TAKEAWAY

- The #AvengersEndgame Twitter conversation was primarily driven by **retweets and content amplification**.
- Original content creation represented a smaller portion of the observed activity.
- The approximately **90% retweet ratio** indicates strong sharing behavior.
- Overall sentiment was predominantly positive.
- Positive sentiment was further amplified through retweets.
- Original tweets showed more neutral and negative sentiment compared with the overall dataset.

SOURCE : ..\instagram.ipynb.pdf , ..\Instagram_Social_Media_Analytics_Report.pdf , ..\Meeting 2 - Twitter Dataset Analysis (EDA).pdf , ..\Twitter Dataset Report.pdf

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1. MEETING OVERVIEW

The meeting focused on converting social-media data into measurable signals for the SIH prototype. The discussion covered Instagram time and trend analysis, content performance, engagement signals, Twitter sentiment analysis, TF-IDF, Logistic Regression, data leakage prevention, model tuning, and the next technical steps.

MLI TOPICS DISCUSSED

- Instagram dataset structure and data validation
- Posting-time and day-of-week analysis
- Daily posting, engagement, reach and sharing trends
- Content-category, media-type and hashtag analysis
- Call-to-action (CTA) analysis
- Twitter sentiment-analysis pipeline
- TF-IDF and Logistic Regression
- Preventing data leakage
- Hyperparameter tuning and fair model evaluation
- Future ML, NLP, propagation and dashboard development

MLII 2. INSTAGRAM DATASET FOUNDATION

The current Instagram dataset contains 29,999 posts, 23 columns and an observed post_date span of 366 days. The analysis reported zero missing values in the checked dataset.

- Account fields: account_id, account_type, follower_count
- Content fields: media_type, content_category, traffic_source
- Time fields: post_datetime, post_date, post_hour, day_of_week
- Engagement fields: likes, comments, shares, saves
- Exposure fields: reach and impressions
- Growth field: followers_gained
- Text/metadata fields: caption_length and hashtags_count
- Model target candidate: performance_bucket_label

MLIII 3. TIME AND TREND ANALYSIS

3.1 POSTING VOLUME BY HOUR

Posts were grouped by post_hour to identify when posting activity is concentrated. Posting volume and performance are different signals: the hour with the most posts is not automatically the hour with the best engagement.

3.2 HIGHEST ENGAGEMENT HOURS

The five highest average engagement-rate hours reported in the analysis are:

• 03:00

• 08:00

• 02:00

• 17:00

• 20:00

The differences between these hours are small, so posting time should be treated as one feature among several rather than as a standalone predictor.

3.3 ENGAGEMENT BY DAY

Engagement rate was also averaged by day of week. The presentation reports Sunday, Tuesday and Friday among the higher averages. The differences are close, so day of week should be combined with content, format, account and audience signals.

3.4 REACH AND FOLLOWER GROWTH

- Average reach was calculated for each posting hour to measure exposure.
- Average followers gained was calculated by hour to add an audience-growth signal.
- Reach measures exposure, while followers_gained provides a growth-related signal.

3.5 DAILY TRENDS

- Daily posting volume was calculated from post_date.
- Daily average engagement rate was calculated to observe performance changes.
- Daily average reach was calculated to observe exposure trends.
- Daily total shares were calculated to observe propagation.

Shares are especially useful for the SIH framework because they provide a propagation signal rather than only a passiveattention signal.

MLIV 4. CONTENT ANALYSIS

4.1 CATEGORY DISTRIBUTION

The dataset contains ten relatively balanced content categories:

Category	Posts
Photography	3035
Fashion	3034
Technology	3025
Lifestyle	3017
Food	3010
Fitness	3004
Music	3003
Travel	2968
Beauty	2953
Comedy	2950

4.2 HIGHEST ENGAGEMENT CATEGORIES

- Music – 0.042808
- Fitness – 0.042720
- Fashion – 0.042615

The differences are small. Category is therefore a useful candidate feature, but it should be combined with timing, media format and account-level signals.

4.3 MEDIA TYPE AND HASHTAGS

- Average engagement was compared across media types.
- Average engagement was grouped by hashtag count.
- Hashtag count alone does not show a simple guaranteed increase in engagement; the relationship is noisy.

4.4 CALL-TO-ACTION ANALYSIS

- No CTA: average engagement rate = 0.042232
- CTA: average engagement rate = 0.041875

In this dataset, posts without a CTA have a slightly higher average engagement rate. This is only an association and does not prove that adding a CTA reduces engagement.

MLV 5. FROM ANALYTICS TO THE SIH FRAMEWORK

1. Historical Data – collect data through CSV/API sources.
2. Signals – calculate time, engagement, exposure, growth and content signals.
3. NLP – add sentiment and topic information from raw text.
4. Propagation – study shares and account relationships.
5. Risk Scoring – combine multiple signals into an explainable score.
6. Dashboard – present trends, anomalies, topics and risk signals for human review.

Important: a negative or high-engagement post is not automatically a threat. Multiple signals such as abnormal propagation, rapidly rising topics, sentiment/context and network behaviour should be considered together.

MLVI 6. SENTIMENT ANALYSIS – ML WORK

The sentiment-analysis component uses two Twitter sentiment datasets and follows a complete pipeline from data cleaning to final evaluation.

6.1 DATA PREPARATION

- Training dataset: approximately 74K tweets initially; 57,178 usable tweets after cleaning.
- Testing dataset: approximately 1K tweets initially; 827 after cleaning; 398 genuinely unseen tweets used for final evaluation.
- Irrelevant columns such as platform/topic metadata were removed so the model learns from tweet text.

6.2 PREVENTING DATA LEAKAGE

Data leakage occurs when test/evaluation information is available during training. It makes evaluation scores unreliable and can make a model appear more accurate than it really is.

The initial audit found that around 52% of the provided test tweets also appeared in the training data. These overlapping tweets were removed, leaving 398 tweets that do not appear in the training set.

This creates a fairer evaluation of real-world model performance.

6.3 TF-IDF + LOGISTIC REGRESSION PIPELINE

7. Clean the tweet datasets.
8. Remove irrelevant metadata where appropriate.
9. Check duplicates and train/test overlap.
10. Use TF-IDF to convert tweet text into numerical features.
11. Train Logistic Regression on the TF-IDF features.
12. Tune hyperparameters.
13. Evaluate only on genuinely unseen test tweets.

6.4 WHAT IS TF-IDF?

TF-IDF (Term Frequency–Inverse Document Frequency) converts text into numerical features. It gives useful importance to words that help distinguish documents while reducing the influence of words that appear very frequently across many documents.

6.5 WHAT IS LOGISTIC REGRESSION?

Logistic Regression is a classification algorithm. Here, it learns patterns in the TF-IDF features and uses them to predict the sentiment class of a tweet.

6.6 MODEL RESULTS

- Initial accuracy: 92.96%
- Final accuracy after hyperparameter tuning: 98.74%
- Final evaluation set: 398 unseen tweets
- Misclassified: 5 out of 398 tweets

The remaining errors were mainly described as ambiguous or context-dependent tweets, including news statements, neutral tweets containing positive words, or tweets requiring domain knowledge.

MLVII 7. CURRENT PROJECT STATUS

- Instagram data validation completed
- Time and trend analysis completed
- Content analysis completed
- Visualization work completed
- Twitter sentiment datasets cleaned
- Train/test overlap and leakage checks completed
- TF-IDF + Logistic Regression model built
- Hyperparameter tuning completed
- Final sentiment evaluation completed on 398 unseen tweets

MLVIII 8. DATA GAPS AND ML LIMITATIONS

- Raw Instagram caption text is not available in the current CSV, so direct Instagram NLP cannot yet be performed from this dataset.

- Account-to-account relationship data is missing, limiting network/propagation analysis.
- Demographic attributes are unavailable for audience profiling.
- The current data is historical rather than a live continuous stream.
- Observed associations do not prove causation.
- performance_bucket_label must be checked for target leakage before ML use.
- If engagement_rate is derived from the target/outcome, it should not be used as a predictor of that target.
- Likes, comments, shares and saves may also be post-outcome variables and can cause leakage depending on the prediction point.

MLIX 9. NEXT TECHNICAL STEPS

Correlation Analysis: Measure relationships among reach, impressions, likes, comments, shares, saves, followers gained and engagement rate.

Feature Engineering: Build meaningful rates, ratios, time features and content features.

Machine Learning: Define the prediction point and target, check leakage, then train and evaluate models.

NLP: Add raw captions/hashtags for sentiment and topic detection.

Propagation Analysis: Add account/relationship data to study information flow.

Dashboard: Combine trends, anomalies, topics, sentiment and risk signals in one human-readable interface.

MLX 10. CONCLUSION

The project has moved from raw social-media data to measurable signals. The Instagram analysis establishes a foundation for time, trend, content, engagement, reach, sharing and growth analysis. The sentiment-analysis work adds an NLP/ML component using TF-IDF and Logistic Regression, with particular attention to preventing data leakage and evaluating on genuinely unseen data.

The next phase is to connect these components through correlation analysis, feature engineering, carefully defined ML tasks, NLP, propagation analysis and an explainable dashboard.

(time_and_trends)analysis: (time_and_trends)analysis.pdf
Instagram_Social_Media_Analytics_SIH_ : Instagram_Social_Media_Analytics_SIH_Presentation_v2.pptx
Sentimental Analysis : Meeting 3 - Sentimental Analysis.pdf

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MLXI 1. INTRODUCTION

The Risk Prioritization Prototype analyzes social-media activity and prioritizes posts for further investigation. The current pipeline is: Time/Trend Analysis → Sentiment → Engagement Signals → Risk Score. The current risk score is a prototype heuristic, not a confirmed threat probability.

MLXII 2. OBJECTIVE

The prototype avoids judging a post using only one signal. It combines sentiment, propagation and engagement-related signals to identify posts that may deserve higher review priority.

- Understand what text expresses through sentiment analysis.
- Identify changes in activity through trend analysis.
- Measure content spread using shares and reach.
- Use engagement as an additional signal.
- Combine these signals into a preliminary risk-prioritization score.

3. WHY MULTIPLE SIGNALS?

Negative sentiment does not automatically mean a threat. Neutral content can also become important if it spreads unusually fast or reaches many people. Therefore, sentiment is treated as one signal among several.

MLXIII 4. SENTIMENT ANALYSIS WITH VADER

VADER analyzes each caption and returns neg, neu, pos and compound scores. The compound score ranges from -1 (highly negative) to +1 (highly positive). The compound value is converted into Positive, Negative or Neutral.

- Positive: 13,804
- Negative: 8,547
- Neutral: 7,648

MLXIV 5. DAILY / WEEKLY SENTIMENT TRENDS

Posts are grouped by date and sentiment. The grouped results count posts in each sentiment class, and percentages make weeks with different total post volumes easier to compare.

MLXV 6. SENTIMENT VS ENGAGEMENT

Posts are grouped by sentiment and mean engagement rate, reach and shares are calculated. The observed averages were very similar across the three sentiment classes. Therefore, negative sentiment alone is not enough to identify a highpriority post.

MLXVI 7. HIGH-SHARE NEGATIVE POSTS

The dataset is filtered to negative posts. quantile(0.90) gives the 90th-percentile share threshold, so posts above it are approximately the top 10% of negative posts by shares.

- Example shares: 516
- Example reach: 46,738
- Example engagement rate: 0.2398
- Example sentiment score: -0.8655

These are candidates for review, not confirmed threats.

MLXVII 8. BUILDING THE PROTOTYPE RISK SCORE

The relevant features are normalized to a comparable 0–1 scale.

- Negative Score = strength of negative sentiment.
- Share Score = relative magnitude of shares.
- Reach Score = relative magnitude of reach.
- Engagement Score = relative magnitude of engagement.
- The current weights are preliminary heuristic choices.

MLXVIII 9. RISK SCORE FORMULA

Risk Score = [(0.30 × Negative Score) + (0.25 × Share Score) + (0.25 × Reach Score) + (0.20 × Engagement Score)] × 100

Example: 0.8, 0.7, 0.9 and 0.6 produce a combined score of 0.76, giving a Risk Score of 76.

The weights are manually selected heuristics, not learned by ML. The score is therefore a prioritization signal, not a probability that a threat is real.

10. RISK LEVELS

Risk Score	Risk Level
0–39	Low
40–69	Medium
70–100	High

High means stronger risk-related signals and higher review priority. It does NOT mean confirmed threat.

11. WHAT IS COMPLETE?

- Data cleaning
- Time analysis and trend analysis
- Correlation analysis
- VADER sentiment analysis
- Daily / weekly sentiment trends
- Sentiment vs engagement analysis
- High-share negative-post analysis
- Prototype rule-based risk score

MLXIX 12. NEXT STEP: ACTUAL ML THREAT DETECTION

1. Create a properly defined threat/risk label using expert-reviewed examples.
2. Build meaningful NLP, propagation and network features.
3. Split data into training and testing sets.
4. Train and evaluate a supervised model such as Logistic Regression or Random Forest.

5. Connect predictions to the backend/dashboard with human review.

MLXX 13. OVERALL SYSTEM FLOW

Social-media data → Time/Trend Analysis → Sentiment Analysis → Engagement & Propagation Signals → Risk Score → Risk Level → Human Review

MLXXI 14. KEY TAKEAWAYS

- Sentiment tells us what the text expresses.
- Trend analysis tells us when activity changes.
- Shares, reach and engagement indicate how strongly content is spreading or performing.
- Risk scoring combines these signals to prioritize posts.
- A negative post is not automatically a threat.
- The current score is a heuristic prototype, not a probability.
- The next step is a properly labeled and validated ML threat-detection model.

MLXXII 15. CONCLUSION

The Risk Prioritization Prototype provides an initial framework for combining sentiment, temporal trends, reach, shares and engagement into a single prioritization signal. The current system is rule-based with manually selected weights. Future work will focus on expert-reviewed labels, NLP and network features, supervised ML, validation and dashboard integration.

SOURCE: ..\Social_Media_Analytics_Risk_Scoring_Prototype.pdf